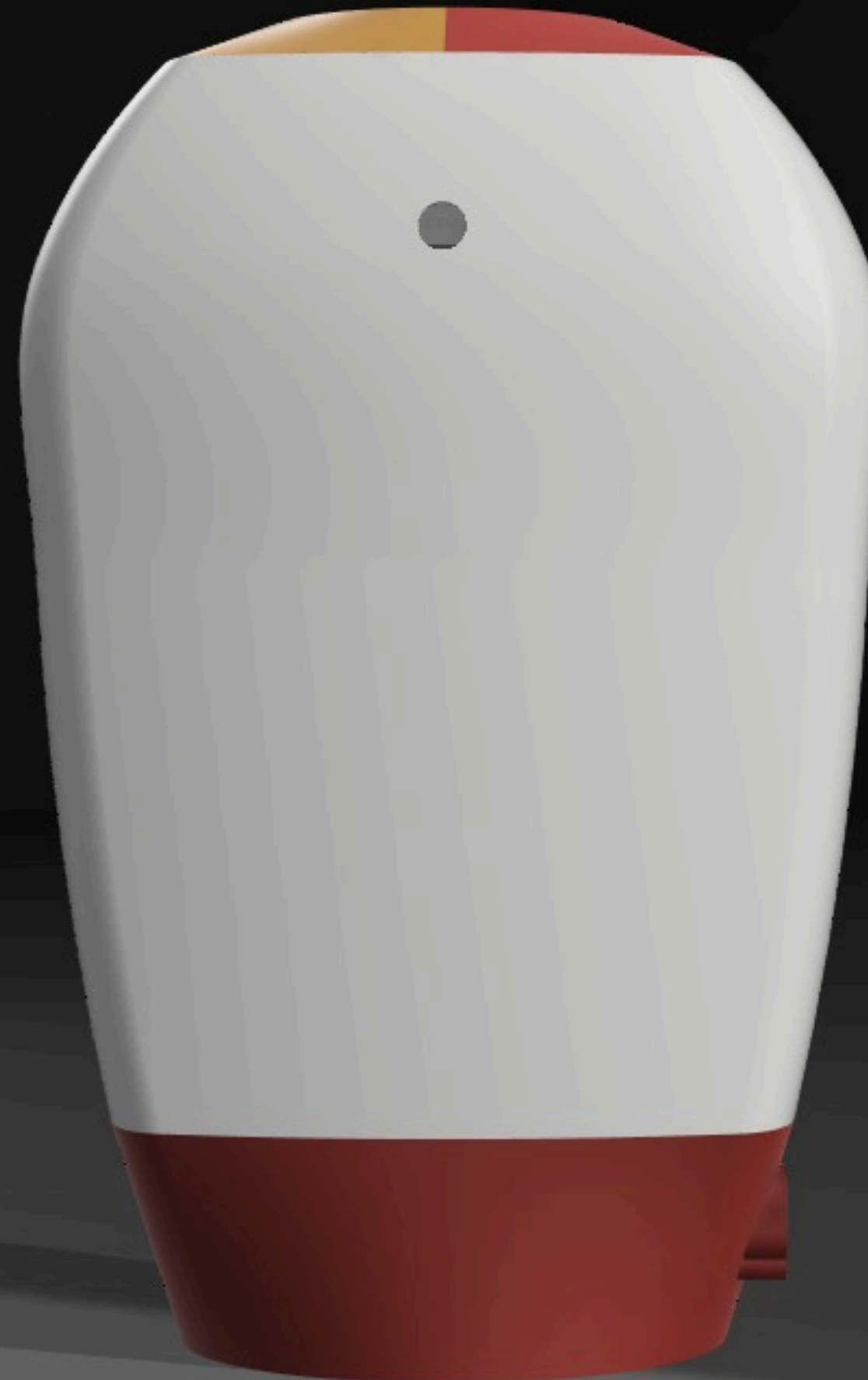


COMPLETED

Seagull

Surf Life Saving: AI Swimmer Detection

A camera with an in-built artificial intelligence detection and tracking software to accurately and automatically locate swimmers in the rip currents to assist in Surf Lifesaving preventative operations.



15

Design Interactions

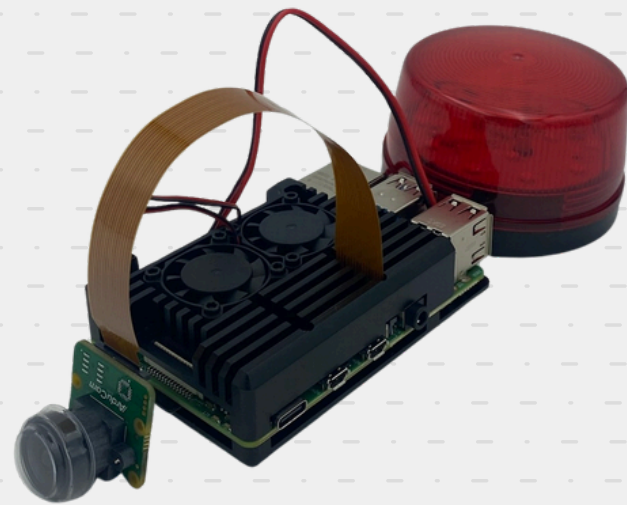
2nd

Place State
ranking

1

Shape Exhibition
Nomination

Design Iterations



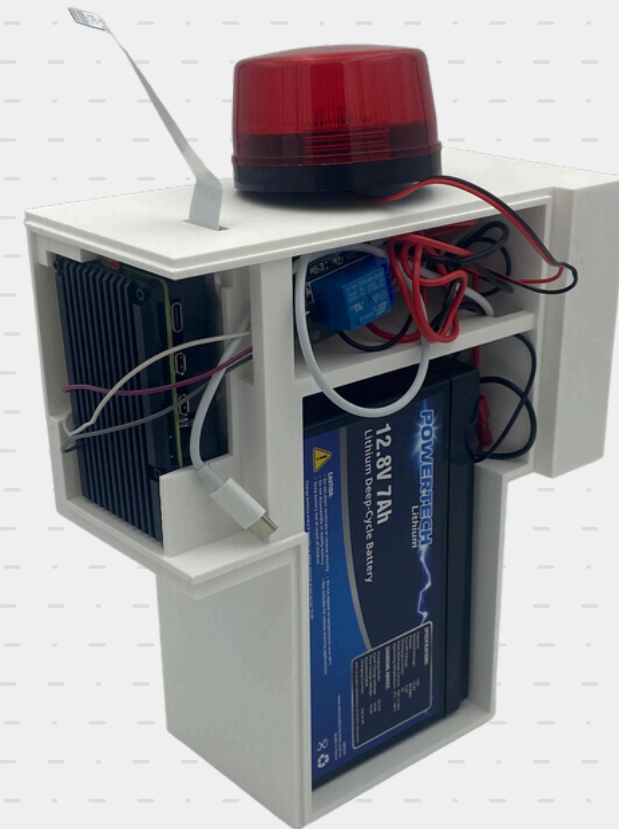
MARK 1

Main electrical components wired with uploaded AI swimmer detection software



MARK 2

Initial Raspberry Pi and Camera dimensions gauged with laser cut plywood housing



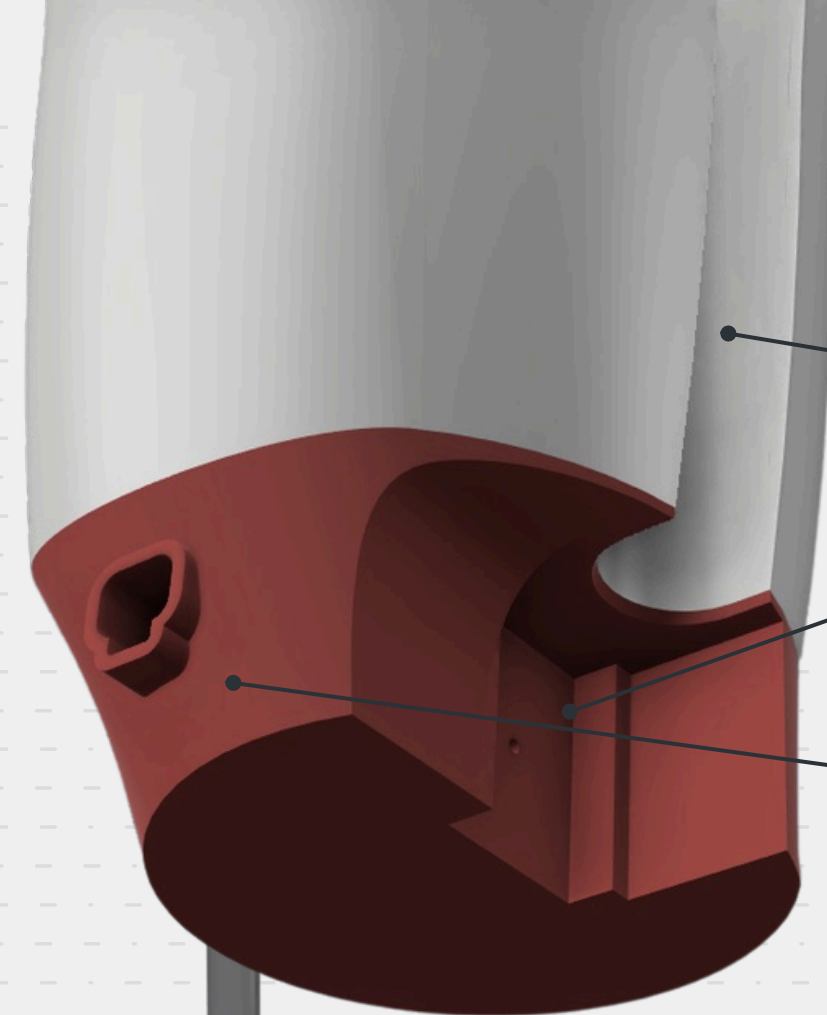
MARK 4

Housing module including battery, relay, transformer with counter balance on right for better balance on pole



MARK 15

Optimised housing module with waterproofing, shock resistant outershell and increased access for maintenance

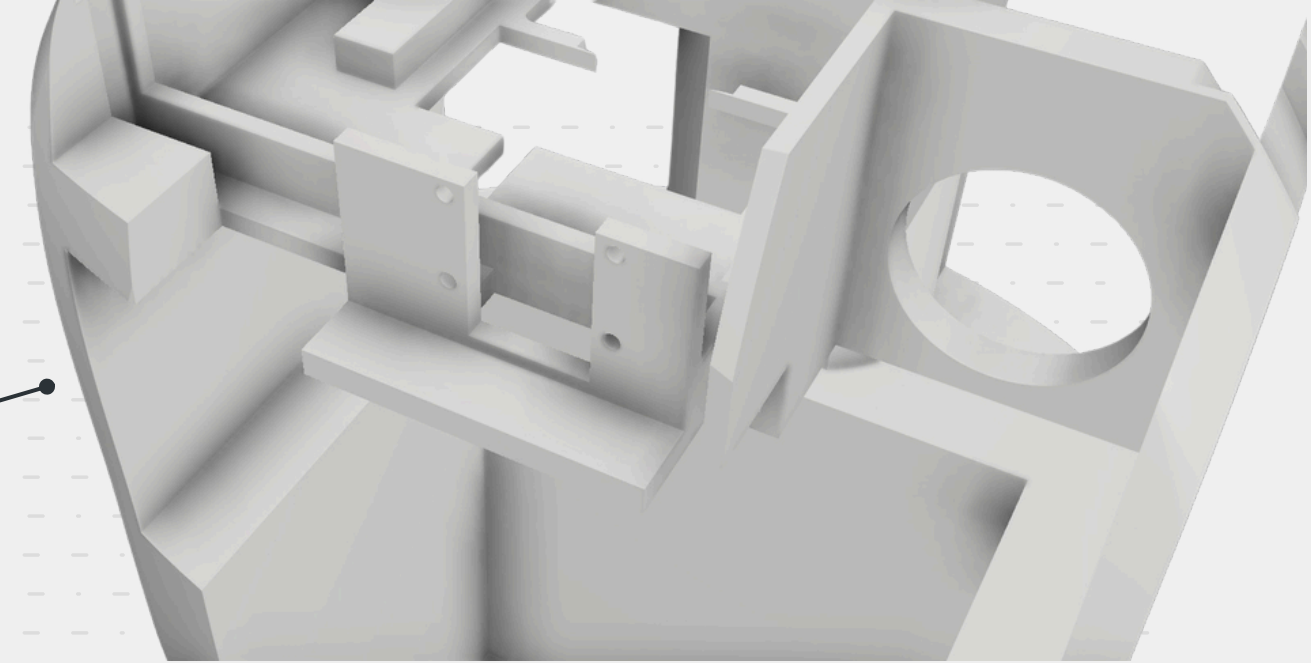


MARK 12

Slot for Surf Lifesaving poles
for mounting stability

Area for GoPro universal
mounting for easy
attachment and operation

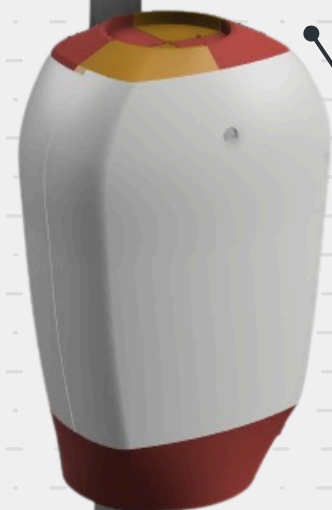
Water proof recharging port
for battery



MARK 7

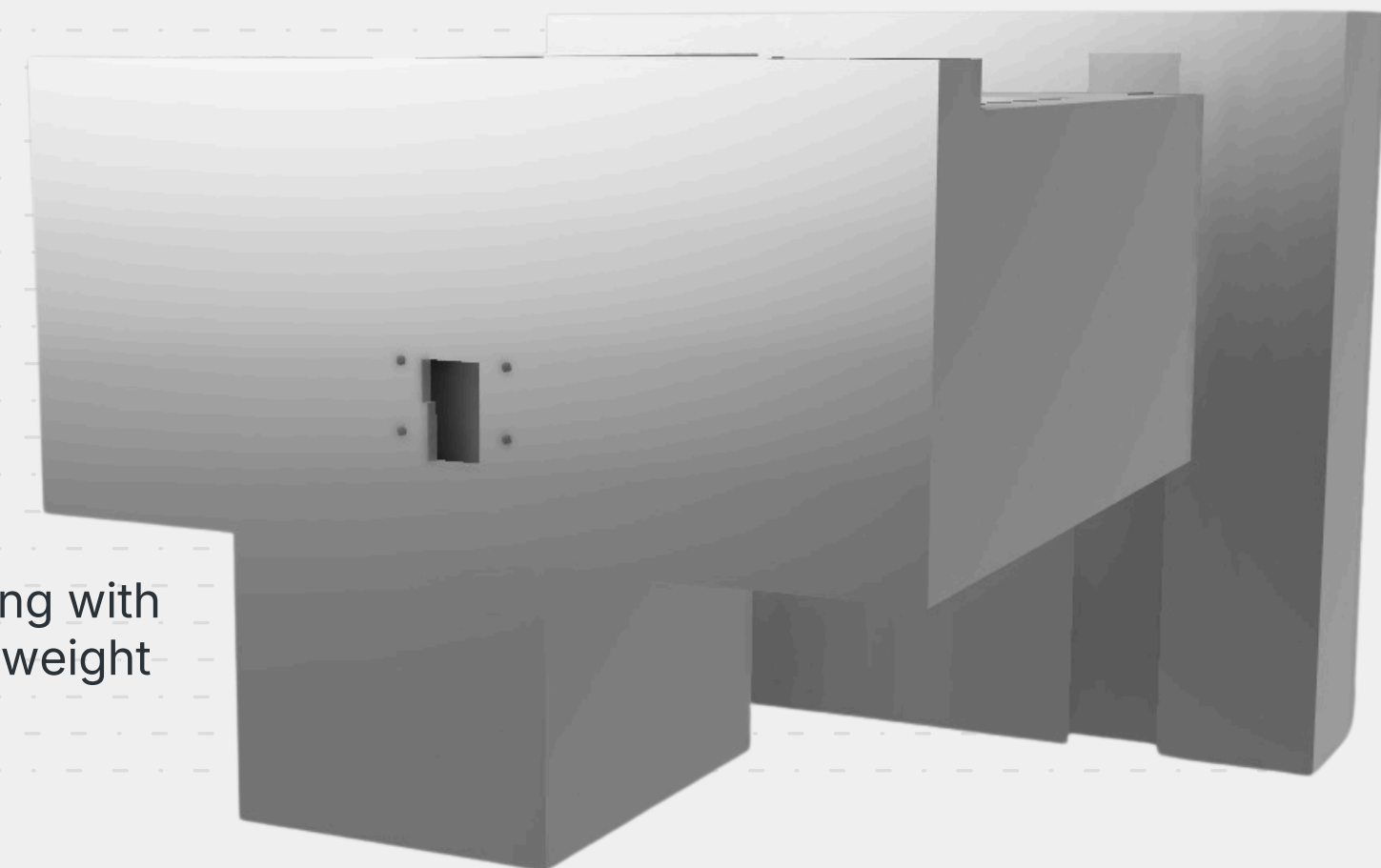
Interior Housing

Fusion 360 Models



MARK 15

Surf Lifesaving branding
colours



MARK 3

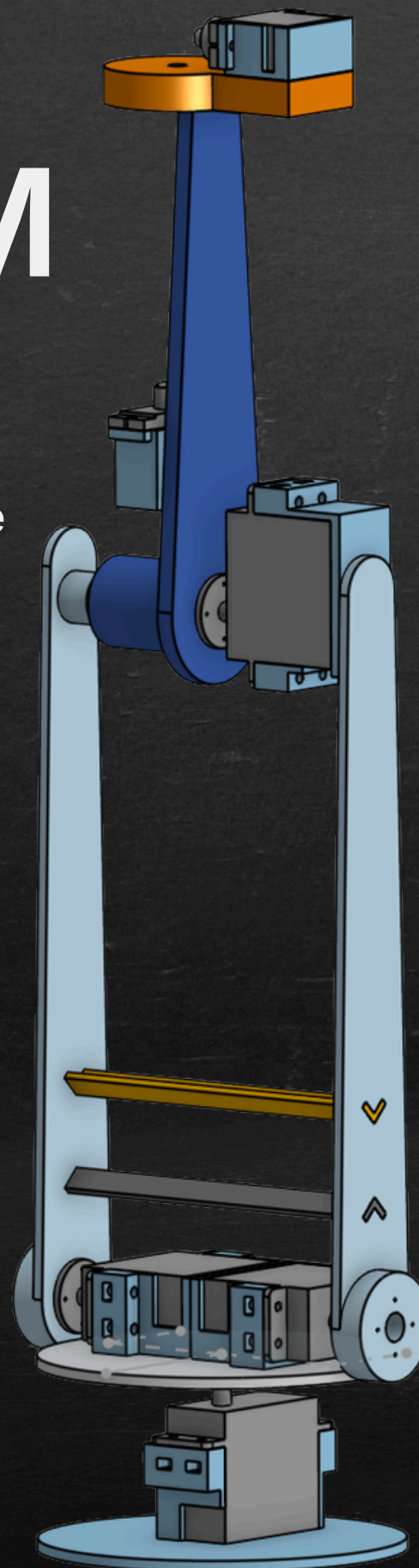
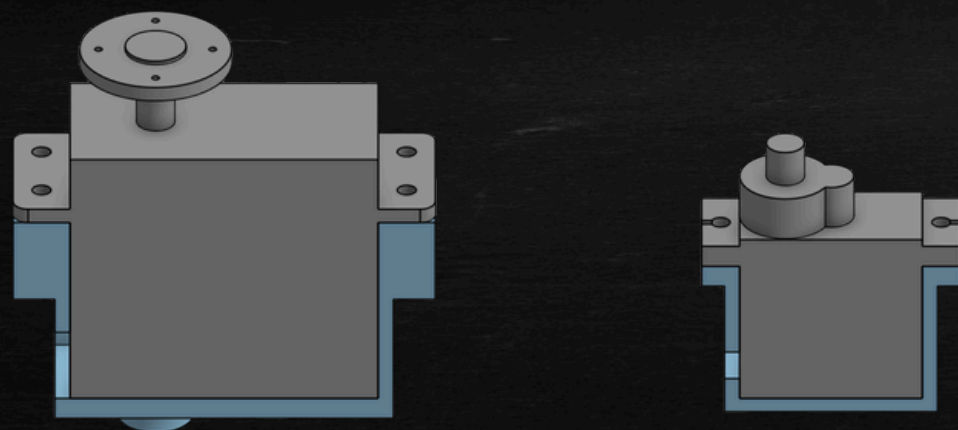
First 3D printed housing with
dovetail back counterweight
to attach to pole

IN PROGRESS

5 DOF ROBOTIC ARM

This robotic arm is to clear completed prints from my 3D printer and neatly place them on my desk. An alternative solution would be to design a simple "sweeper" however this would risk breaking delicate prints or muddle up print bed groups.

This is a self funded hobby project so all mechanical features will be aimed to be solved with 3D printed solutions or salvaged parts from household items and old projects.



Phase I (current phase) - physically building the arm with the MG955 and MG90S servo motors.

Phase II - writing the functioning code and calculating the inverse kinematics on Matlab.

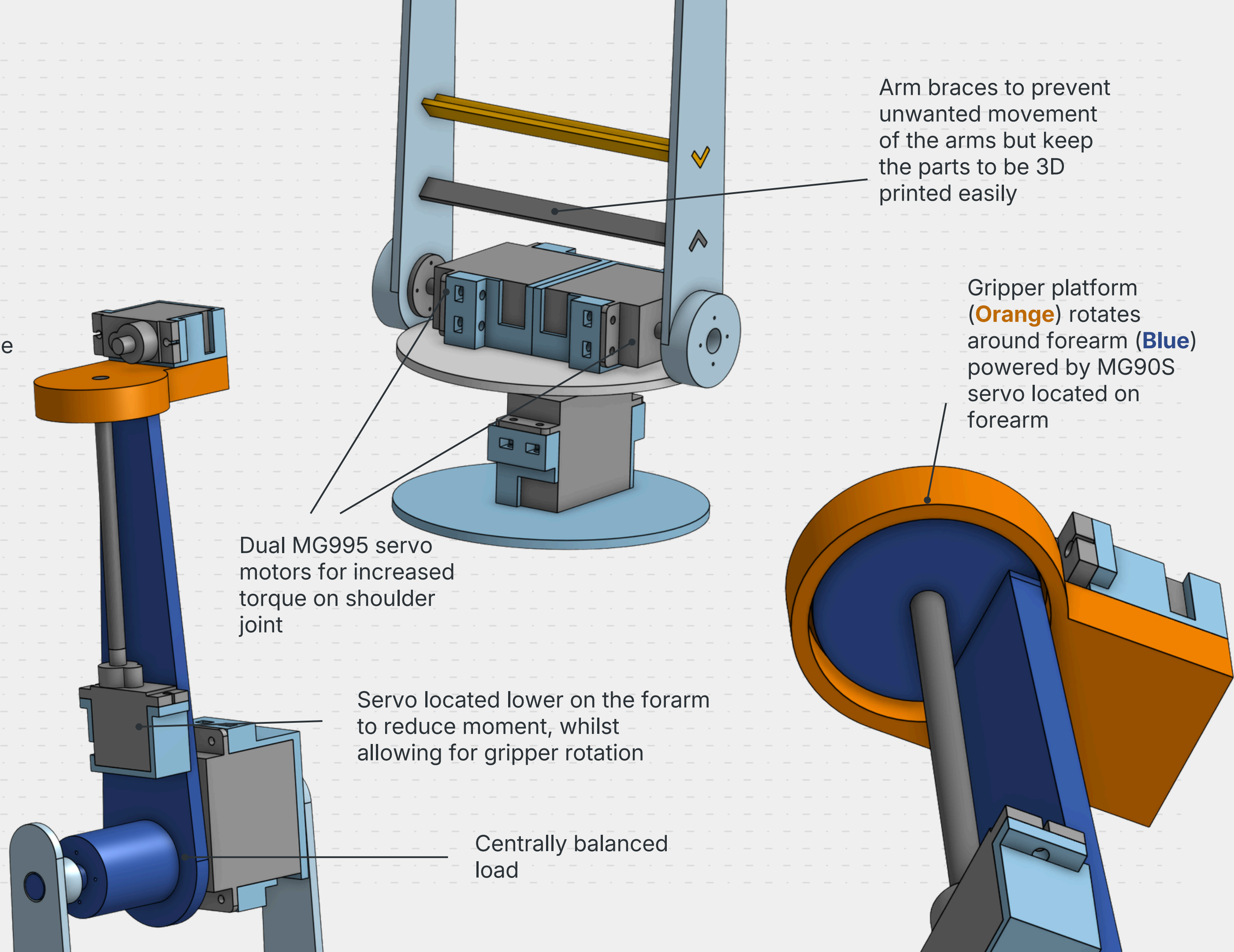
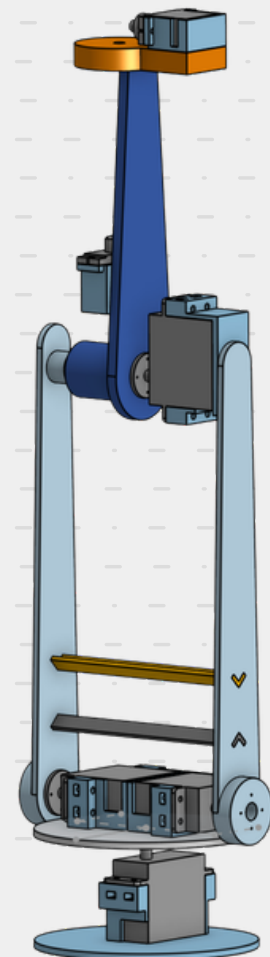
Phase III - adding a gripper as well as a camera and repurposed piezoelectric sensors in the gripper to allow for remote control.

Phase IV - coding an app and UI so that the robot can be controlled remotely via a phone.

MARK I

Servo Orientation

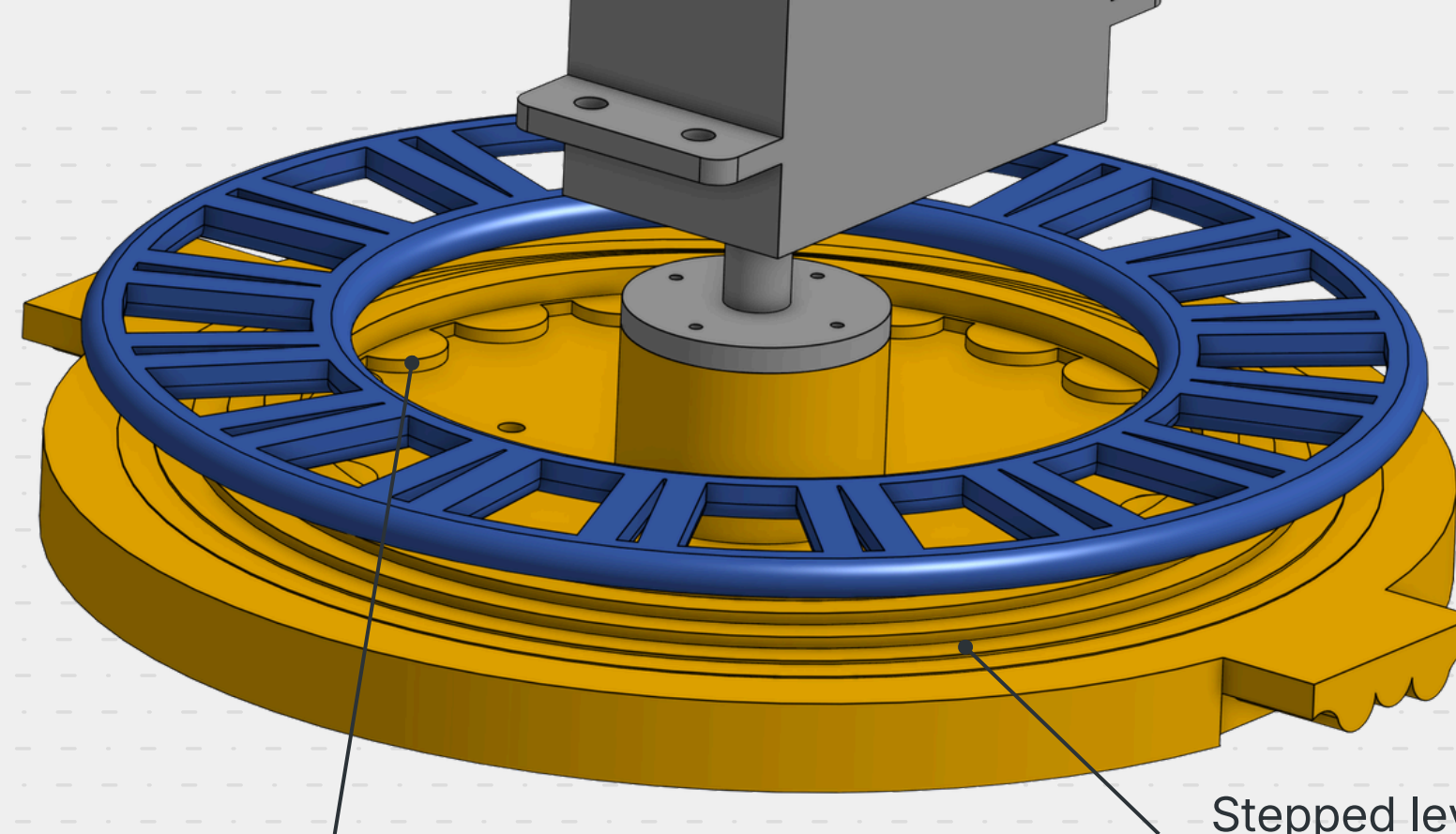
This prototype aimed to work out the servo placement as well as the dimensions of all the arms and joints supported by moment calculations.



MARK 2

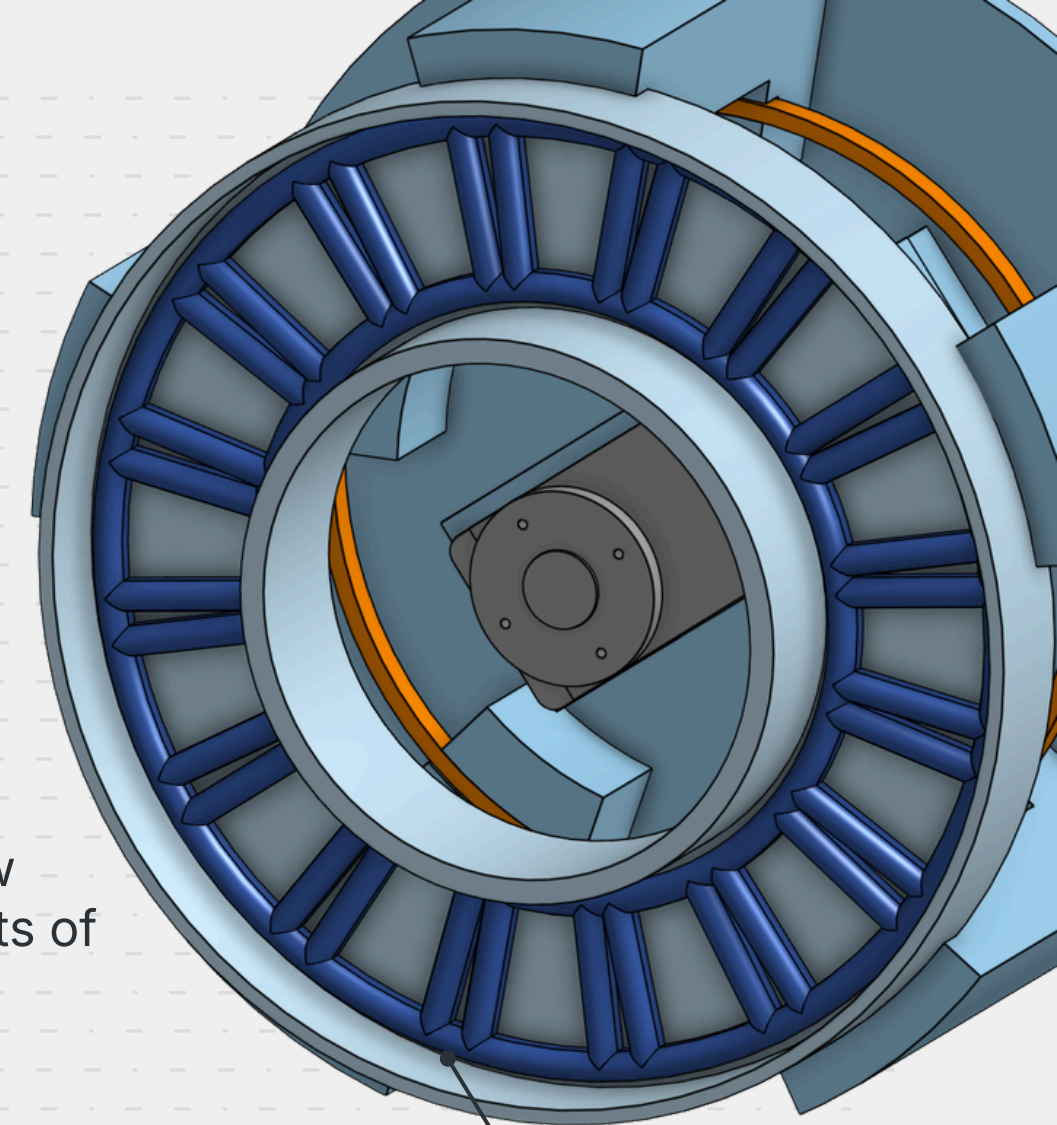
Part Support

This prototype aims to build viable shells and supports to house the servos mapped out in MARK I. Currently only the base and shoulder servo supports have been built.

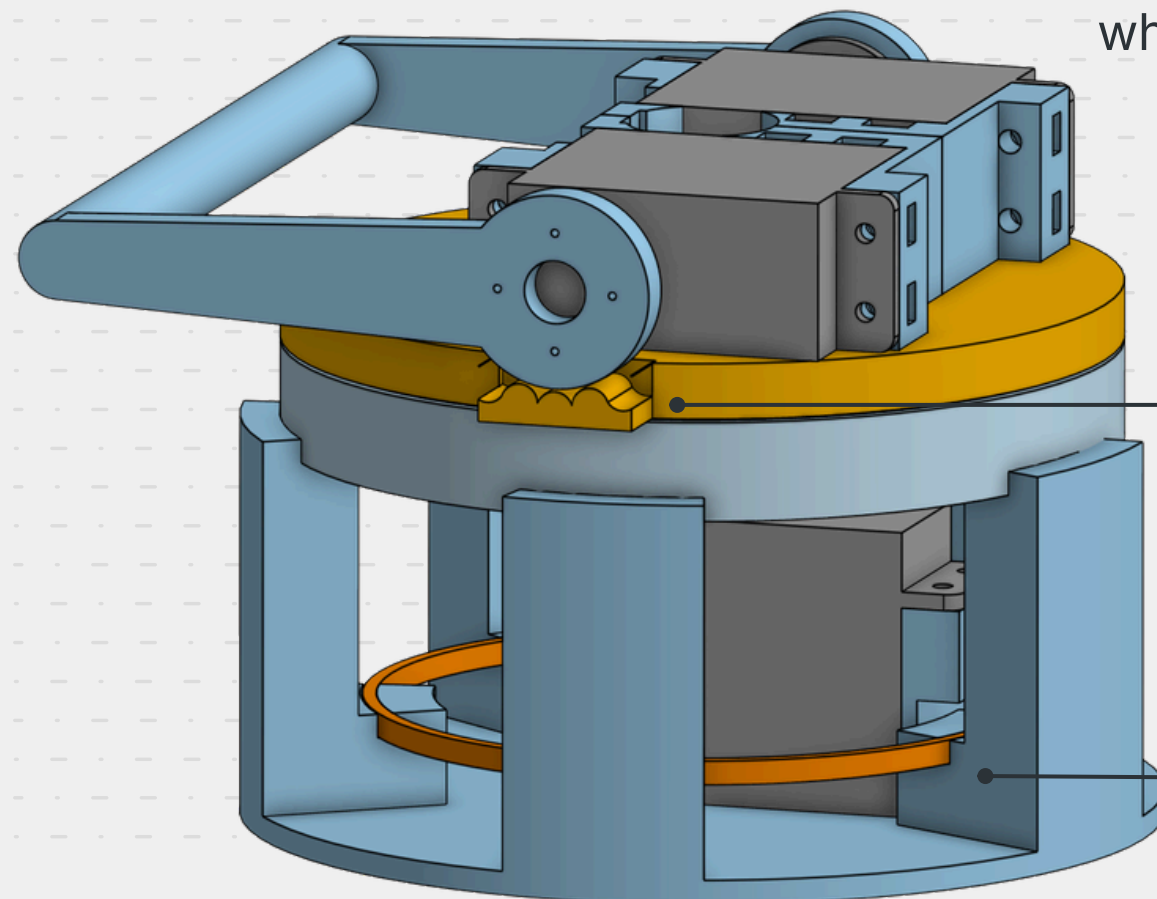


Circular offsets keeps the base (**Orange**) centred on the bearing housing (**Light Blue**) whilst minimising friction

Stepped levels allow base to have 5 points of contact with each truncated bearing

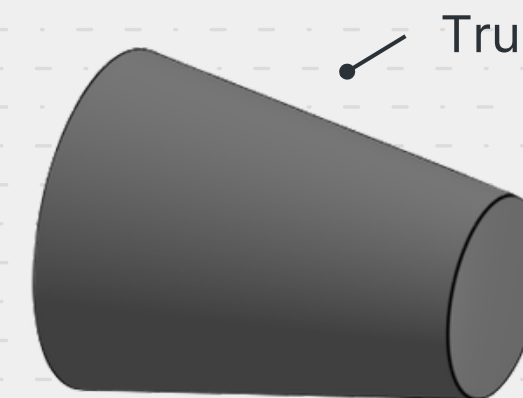


Bearing guide rail (**Blue**) keeps the truncated cone bearings in place



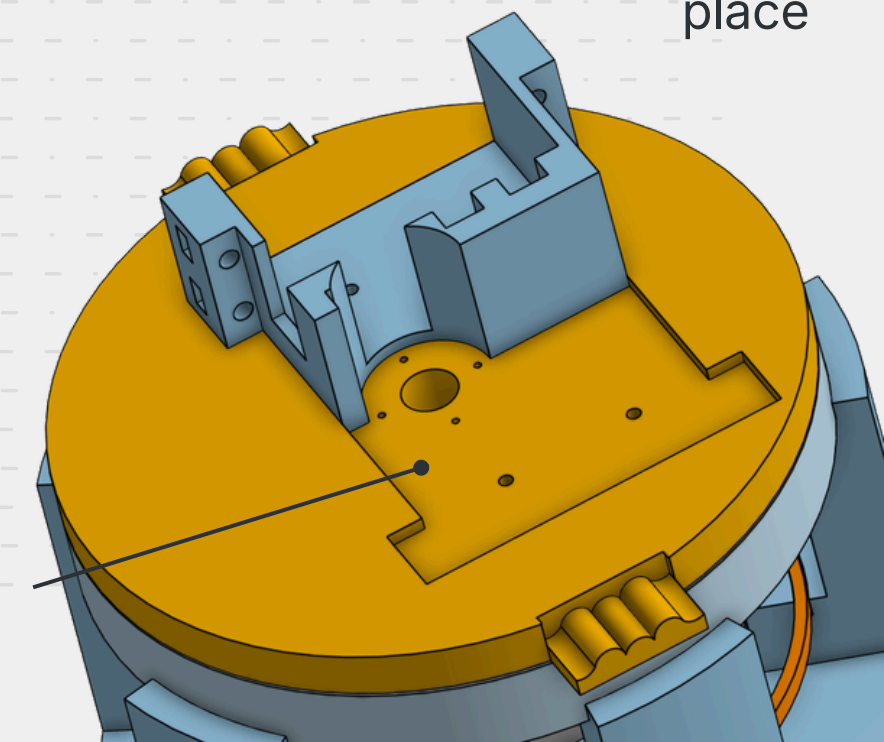
Upper arm support reduces shear forces on servo shaft. circular design minimises contact points and friction, whilst allowing smooth movement

Support ring to brace the support columns from internal forces



Truncated Bearing

Servo housing indent on base allows for easy alignment whilst minimising 3D printing supports when printing the base



COMPLETED

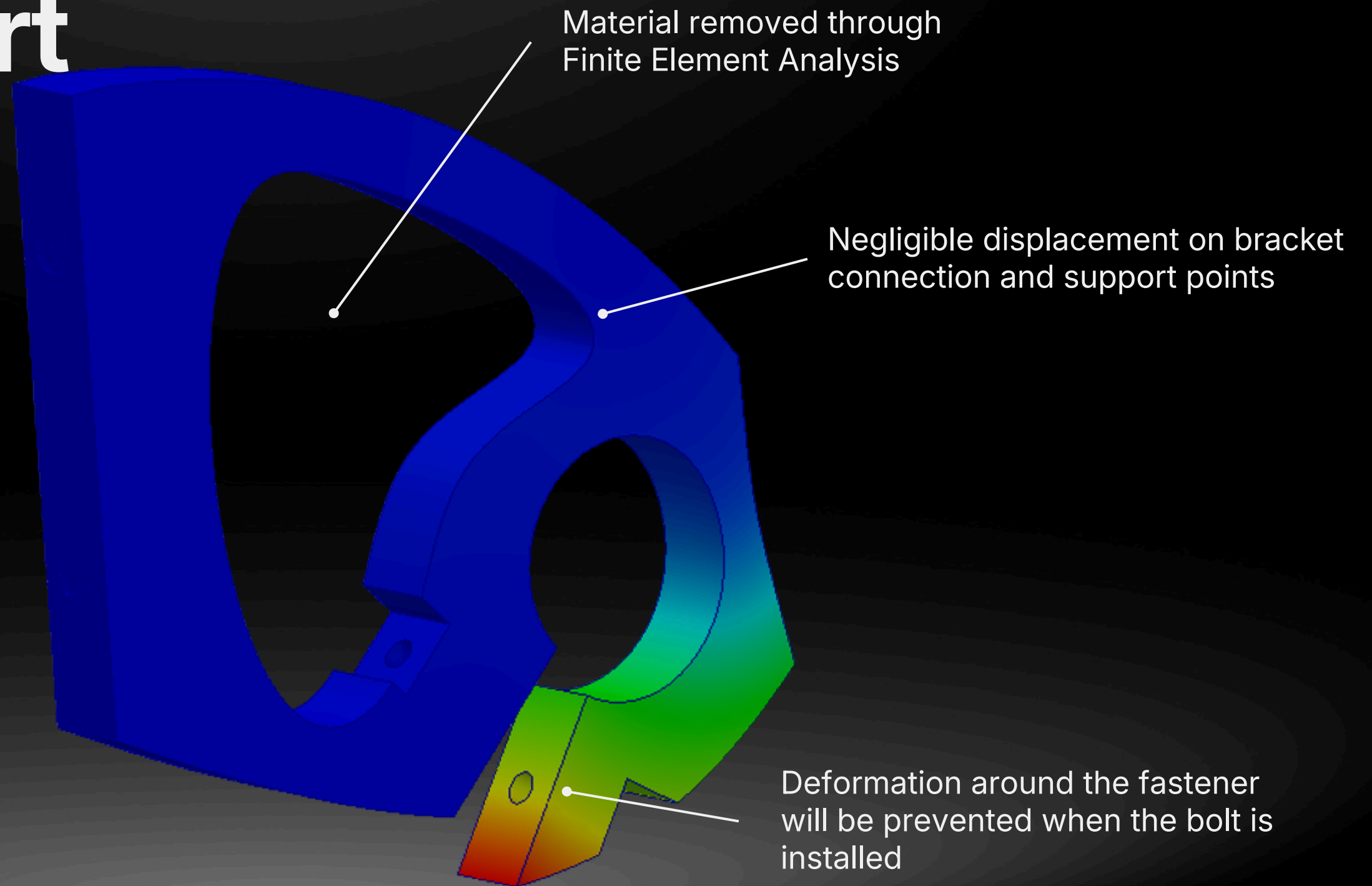
USYD Motorsport

FoS and Displacement simulations

This bracket was designed as a task for the USYD Motorsport Structures team to mount the jacking bar and withstand a 700N lifting load.

This bracket provides no advantages in performance however is regulated by FSAE competition rules. Thus, should minimal displacement and minimal weight.

This bracket was designed using an interactive process of FEA simulations then testing the changes with FoS and Displacement simulations.

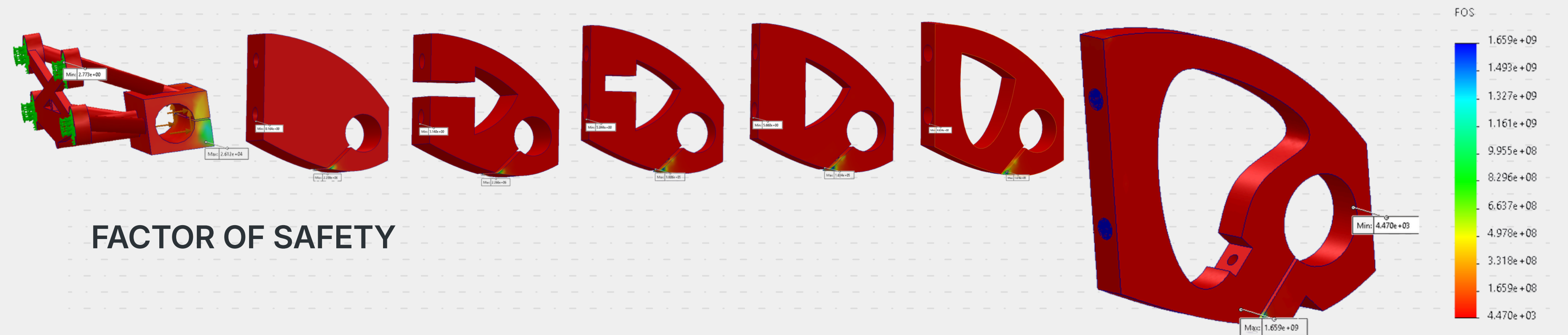


Solidworks Simulations

DISPLACEMENT



FACTOR OF SAFETY



COMPLETED

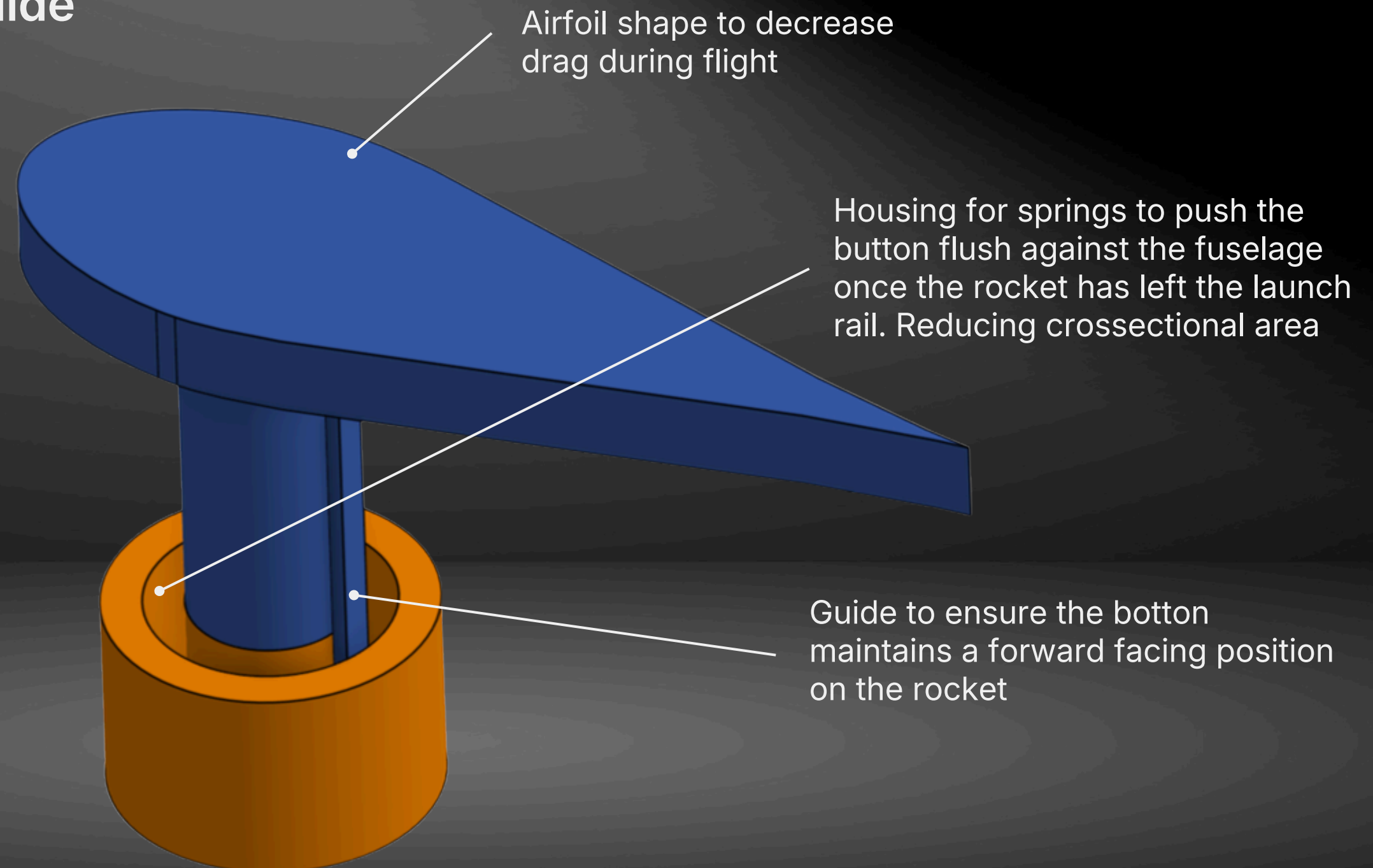
USYD Rocketry

Aerodynamic Retractable Rail Guide

This rail guide button was designed as a task for the USYD Rocketry team.

A rocketry rail guide keeps a model or high-power rocket securely attached to a metal launch rail as it lifts off, ensuring it flies straight until it gains enough speed for its fins to work.

Issues with existing rocketry rail guides is that they cause considerable drag during flight limiting the rockets max apogee.



IN PROGRESS

BeerBound

IOS App for Tracking Nights out

BeerBound is an iOS app, coded on Swift, for tracking nights out with live GPS route recording, automatic stop detection, and photo to location matching using EXIF GPS. This app is currently in review by Apple before it is released to the App Store.

The app's venue, deal, and event data is pulled with API flows from an external database created on Supabase, with offline caching for network failures.

The core engineering challenge was reliability under iOS's limits on running background processes. This included coded logic to ensure GPS fixes are filtered for accuracy, jitter, and long distance jumps caused by poor network connection on public transport and dense urban areas.



Built by our friends,
made for yours.



You forgot?
BeerBound didn't.

